Technology Stack

Date	24 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Technology Stack The technology stack for OptiCrop has been carefully selected to support the development of an intelligent, scalable, and user-friendly crop recommendation system. It includes frontend technologies for creating an interactive user interface, backend technologies for handling application logic and machine learning integration, data storage solutions for managing datasets and trained models, deployment platforms for online accessibility, and development tools.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2 Templates	Chosen to create a responsive, user-friendly, and interactive interface that allows farmers to easily input soil parameters and view crop recommendations across different devices.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python 3.11, Flask 2.2.2	Flask provides a lightweight and efficient framework for handling user requests, integrating machine learning models, and generating realtime crop recommendations.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	Agricultural Dataset (CSV), Pickle Files (.pkl)	The dataset stores crop and soil information, while Pickle files store the trained Random Forest model and scaler for efficient prediction processing.

4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Render, Procfile	Render offers a simple and costeffective cloud deployment platform with GitHub integration and automatic deployment capabilities.

5	**Version Control & CI/CD**	Git, GitHub	Git and GitHub enable source code management, version tracking, team collaboration, and continuous project updates.
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	These tools support machine learning model development, data preprocessing, visualization, exploratory data analysis, and model evaluation within the OptiCrop system.